2. Product safety overview

2.1. General principles

Spot is a legged robot capable of mobility on a variety of terrains. Spot uses multiple sensors and three motors in each leg to navigate in indoor and outdoor environments, maintain balance and attain postures. Spot is capable of carrying and powering attachments.

> **CAUTION**
>
> Spot's behavior while in motion is dynamic and can be subject to local conditions that may generate unexpected motion.

2.2. Intended use

Spot is intended for the professional use of its locomotion and carrying capabilities in either industrial, restricted, or supervised environments.

Spot is intended to be used in dedicated areas, where its possible presence is clearly demarcated or indicated. The limits of locomotion and environments are detailed in Restrictions on the environment.

Spot can carry attachments intended for sensing, monitoring, or other non-contact interaction. Other uses are not intended. The integration of an attachment could in fact result in additional hazards or different risks when the attachment compromizes the stability of Spot, includes dangerous materials, or creates hazardous emissions of all types. For the limits and conditions in which the integration of an attachment can no longer be considered within the intended use, see Integrate attachments.

Spot is not intended for uses involving concurrent human-robot activities sharing the same space or the same equipment to complete an assigned task/mission (sometimes known as "collaborative applications"). People can be occasionally present in the same space as Spot for purposes unrelated to Spot operations (see "Bystander" in Key terminology). The limits of occasional exposure are detailed in Risk assessment.

Any use outside of the intended use is subject to a risk assessment and reduction by the final user. Boston Dynamics recommends that you perform a specific risk assessment to verify:

- the finalized layout of deployment, the setup, and any training activity dedicated to illustrate Spot's working zones;
- the verification of the conditions of the environment, with particular attention to obstacles affecting Spot's navigation that may become apparent after the initial commissioning;
- any additional safeguards or other risk reduction measures;
- potential unplanned maintenance and troubleshooting.

The risk assessment for Spot considered reasonably foreseeable misuses. Explicitly prohibited uses include:

- Underwater and airborne applications.
- Use as a weapon or to enable any weapon.
- Any use as – or enabling the use of – a Certified Medical Device. Access and operation in healthcare facilities subject to limitations.
- Use in personal care, medical treatment or life-critical applications.
- Use in home environments.
- Transportation of persons or animals.
- Transportation of hazardous materials or substances.
- Intentionally harming any person with Spot or by using attachments mounted on Spot.
- Use for any illegal purpose.
- Use as a climbing aid.
- Interfering with Spot's sensors so as to impair their functioning, intentionally altering the environmental conditions so as to impair the functioning of Spot's sensors, or intentionally altering the environmental conditions so as to impair Spot's locomotion.

DANGER

Any misuse of Spot can potentially cause severe personal injuries or result in significant material hazards.

2.3. Modes of operation

Spot can be operated in various modes.

- **Manual:** Direct operation of Spot by displaying images from Spot's cameras on a remote controller. See Remote controllers and Drive Spot with remote control. All Spot operations are supervised and executed by a human driver who is responsible for verifying the surrounding conditions.
- **Automatic:** Autowalk missions can be recorded and replayed by Spot. During replay Spot operates automatically. See Automatic operation.

The frequency and duration of Spot operations in either manual or automatic modes varies greatly depending on the specific use of Spot.

For example, foreseeable uses in applications including routine inspections of industrial assets can be configured along a route in manual mode and then be repeated and completed largely in automatic mode. While operating in automatic mode, Spot is expected to encounter bystanders not involved in its operations for very brief moments along routes. While operating in manual mode, interactions with bystanders are expected to be longer, although limited in time, and may include people attending to their own tasks in the operating environment.

A complete analysis of foreseeable exposure to Spot is reported in Risk assessment.

CAUTION

Spot should always be remotely controlled by properly trained Operators, or execute automatic operations configured or programmed by a trained professional.

Failure to properly verify programmed applications may result in unexpected hazards during operations.

2.4. Locomotion primitives: gaits and specialized modes

2.4.1. Walk ("trot")

Spot's default gait. Spot moves at variable speed on alternating pairs of legs (front-right/hind-left, front-left/hind-right).

On a firm flat surface such as concrete, wood floor, or thin carpet, Spot's movement while walking is a regular cadence of each pair of legs swinging and touching the ground together. The body remains approximately stable at the same height and attitude.

Spot will change body posture, the pose of the legs, and gait pattern when necessary to perform programmed actions (e.g. pointing an attachment in a given direction) or to maintain balance under disturbances.

2.4.2. Crawl

Spot moves slower than trot on flat terrain and keeps three feet on the ground at all times.

2.4.3. Quick-step slip prevention

In the Walk gait, Spot may take extremely rapid steps while recovering from a slip. This greatly reduces the chance of falls on ice, wet or oily floors, and other slippery surfaces.

2.4.4. Stair mode

During both manual and automatic operations, when Spot detects a stairway in its path, it will automatically adjust its gait to a moderate speed and pitch its body to adjust to the slope of the stairway. Spot will automatically align itself with the center of the stairway to avoid colliding with walls or railings.

Stair locomotion involves sophisticated control and use of sensors. For the best and most reliable performance in these conditions, keep Spot's software fully updated.

WARNING

Failures on stairs cannot be completely eliminated. When using Spot in manual mode, the best results are obtained when operators do not interfere with the automatic stair behavior by suspending, altering, or making rapid changes in manual commands. Actively stopping Spot on stairs has negative effects on stability (see Hazards associated with stopping or other powerless motion).

For more information about driving Spot on stairs, see Navigate stairs.

2.5. Stopping functions

2.5.1. Operational stop

Spot monitors its sensors and can automatically pause movement or remove power from motors in certain situations:

- Signal loss: After 3 seconds without controller communication, Spot will sit. After 8 seconds without communication, Spot will turn off its motors. See Enable AutoReturn to recover from loss of connection to the controller and Autowalk replay supervision.
- Fall detection: When Spot detects a fall, motors are immediately de-energized. Legs will not actively flail or remain stiff under contact.
- Low battery: When Spot's batteries reach critically low levels, Spot will sit and turn off its motors.
- Controller input: Commands to stop or de-energize Spot can be sent from a robot control device. See Remote controllers and Stop Spot.

WARNING

If Spot is on stairs when an operational stop would occur, it will attempt to exit the staircase before performing the operational stop. This may result in automatic locomotion up or down the staircase, overriding other locomotion commands.

Do not stand on stairs, below open rails, or within 2 meters of the bottom of a staircase where Spot is active.

NOTICE

Spot can detect falls that unexpectedly occur. De-energization of legs during falls minimizes damage from the fall and prevents new hazards or bigger risks, such as further stumbling. This behavior is called "smart freeze."

2.5.2. Safety-related stop

Spot can be stopped by interfacing an external safe input signal to the payload port.

The stopping function will result in a de-energization of all motors (Stop Cat. 0 EN IEC 60204-1).

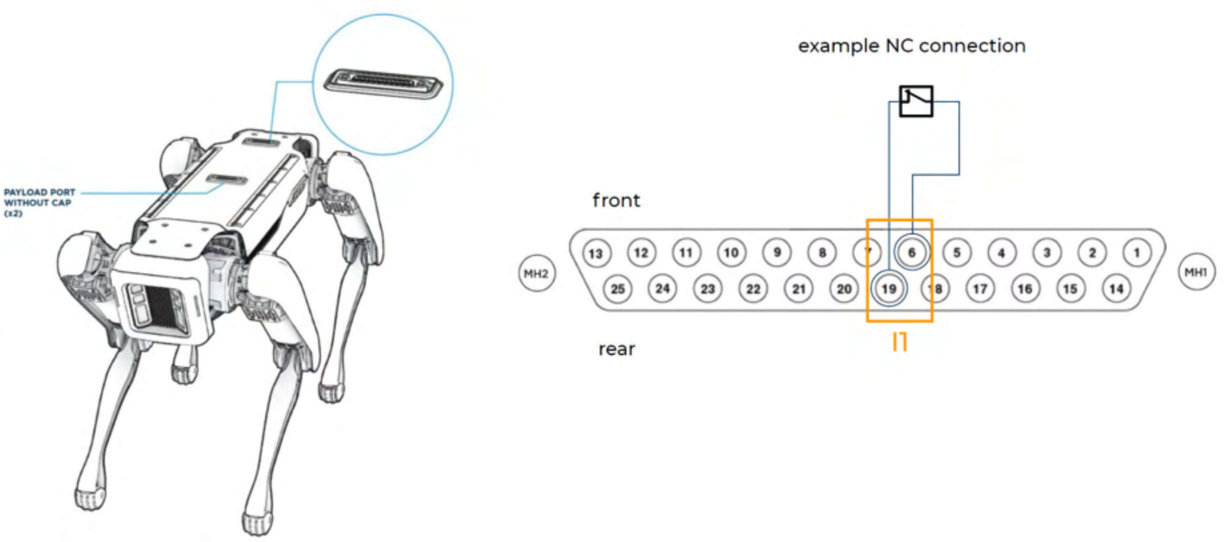
WARNING

When motors are de-energized, Spot will lose its ability to stand and balance. On flat ground, Spot will lower its body. On inclined surfaces or stairs, Spot may tip over.

The maximum response time for the stopping function is 200 ms.

Safe inputs for activating and triggering the stopping function are located on the payload ports as follows:

Item	Description
Safe input location (See figure below)	I1 = pin 19
Interface	Use only one port, either front or rear. See example in figure below. Spot will not work without a cap or a properly configured attachment connected to each port. When a cap is used, the stopping function is disabled.
Safe input default connection	6-19 pair normally closed (NC).
Type of stop	Stop Category 0 (IEC 60204-1) when contact on I1 is opened.
Stopping safety function	Implemented in accordance with ISO 13849-1:2023, of Category 1 and performance level c. The PFHD is 1.1E-6/hr.
Reset	6-19 pair is closed.
Restart	Enable power following commands on the manual controller or issued from programs.
Environment	The safety-related part of the control system operates within the same environment limits established for Spot.



Payload port external signal interface for protective stop.

> **WARNING**
> Use safety-related inputs only for connection to safeguards or emergency stop devices. Do not interface with non safety-related signals.

2.5.2.1. Stopping distance

On flat ground, a safety-related stop will result in a maximum stopping distance of 1 meter. On uneven or sloped surfaces, the stopping distance could be augmented by Spot's motion under gravity and will depend on factors such as the slope or unevenness of the walking surface. For more information, see Hazards associated with stopping or other powerless motion.

2.5.3. Emergency stop (E-Stop)

> **NOTICE**
> The E-Stop button applies only to Spot model number 04-00143531-401. Information about additional Emergency Stop devices applies to all models.

Spot's E-Stop button. Press firmly to activate; twist clockwise to release (unlatch).

Spot includes an Emergency Stop (E-Stop) button located at the top rear corner of its body.

When pressed, the E-Stop button triggers the Safety-related stop, resulting in an Emergency Stop function compliant with ISO 13850. Spot will not resume operations until a deliberate reset and restart procedure is initiated (see Restarting after a stop using the tablet controller).

Spot can be optionally fitted with additional Emergency Stop devices that comply with the same standard. Any additional Emergency Stop input device must meet the requirements of the safe input interface on the payload port as described in Safety-related stop.

The Emergency Stop function is NOT a primary protective measure in any situation, and is manually activated. It should be used only when an immediate emergency cannot otherwise be resolved, for instance in case of accidental entrapment or insufficient clearances.

WARNING

The Emergency Stop function must be used only when you have full visibility of Spot and its surroundings. A manually activated Emergency Stop will override any active control, and therefore could determine additional hazards if it causes a forced loss of stability (see also Hazards associated with stopping or other powerless motion).

CAUTION

Always try to anticipate and escape entrapment conditions.

Do not activate the Emergency Stop function for routine interventions or as a means for a normal stop.

2.6. Velocity limitation

When operating Spot manually in the Walk gait, Spot's maximum speed is adjustable up to 1.6 m/s. To switch between the following maximum speed settings, see Tablet controls (Drive mode).

Speed setting	Velocity
Fast	1.6 m/s
Med (default)	0.9 m/s
Slow	0.5 m/s

CAUTION

Changes to speed settings take effect immediately. Spot may change speed mid-stride.

Speed settings limit the maximum speed that can be commanded by the controller. Spot's actual speed depends on a variety of factors including:

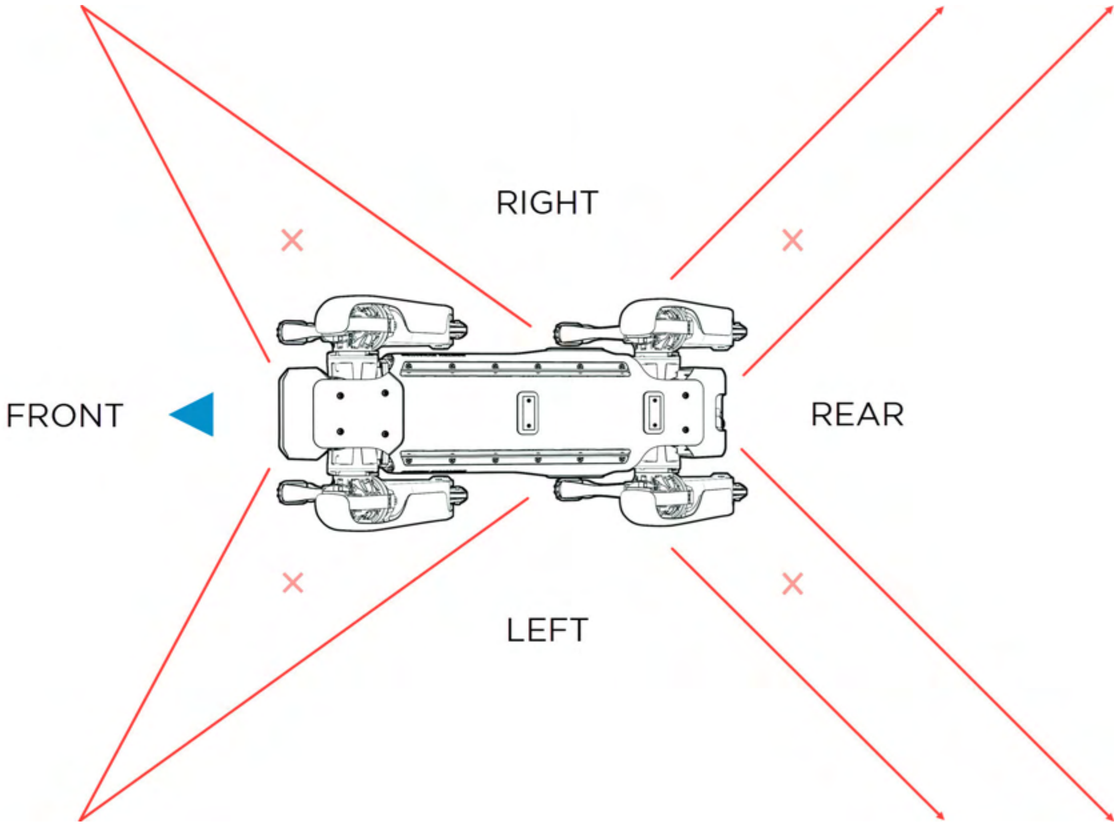
- The current speed setting.
- The current Ground Friction setting (see Platform status panel).
- Variable inputs from locomotion controls, such as the amount of tilt applied to the joysticks on the tablet controller.
- Disturbances that cause Spot to alter its gait so as to maintain balance (see Quick-step slip prevention).
- Disturbances or hardware failures that result in a fall, during which Spot's body or legs may briefly exceed the maximum speed before being de-energized and/or coming to rest.

On a firm flat surface such as concrete, wood floor, or thin carpet, most adults can easily outpace Spot at its default walking speed and there is limited kinetic energy associated with Spot's forward motion.

2.7. Obstacle avoidance

Spot uses its perception system to automatically avoid collisions with obstacles. The perception system consists of five depth cameras: two at the front, one at the rear, and one on each side of Spot.

> **NOTICE**
>
> Although the camera images displayed to an operator may show an overlapping field of view, the perception system has gaps, especially at the rear corners of Spot.



Approximation of gaps in the obstacle avoidance field of view.

Obstacles are mapped and remembered even when Spot's movement brings the obstacle into one of the gaps. However, Spot may fail to detect:

- Moving obstacles.
- Obstacles that are hard to detect until Spot is very close. (For details on the limits of the perception system, refer to Navigational conditions.)
- Obstacles that remain in a gap in Spot's field of view during its entire approach path.

By default, Spot tries to keep a minimum distance of about 7.5 cm between itself and nearby obstacles. You can set an additional cushion of up to 50 cm using the controls described in Drive Spot with remote control. The obstacle avoidance cushion may prevent Spot from traversing doorways and other confined spaces.

CAUTION

Spot may collide with people or objects, even with its obstacle detection system enabled.

Operators and bystanders should assume that Spot may move unexpectedly at any time.

2.8. Auditory and visual (A/V) warning system

NOTICE

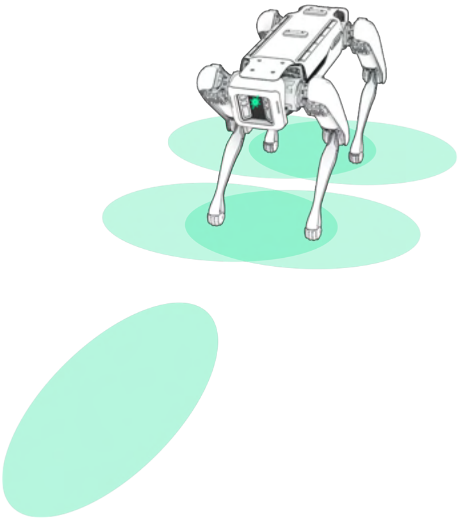
This information applies only to Spot model numbers 04-00143531-401, 04-00143531-601, and 04-00143531-611. To find your model number, check the label inside Spot's battery compartment.

Spot is equipped with an auditory and visual (A/V) warning system to alert nearby people of Spot's presence and operational status.

The warning system consists of five indicators that project colored light onto the ground to the front and sides of Spot's body, a buzzer, and a speaker. When the warning system is enabled, signal patterns activate automatically when specific operating conditions are met (see Warning system light and sound patterns).

The warning system is disabled by default. For instructions on enabling and configuring it, see A/V warning system settings. The most recent configuration will persist across operating sessions and reboots.

2.8.1. Indicators



Areas illuminated by the indicators.

Each indicator can emit multiple colors. Brightness is adjustable and automatically adapts to ambient light conditions.

The indicators are located:

- On the front body panel (1 indicator).
- On the bottom body panel between the front hips (2 indicators).
- On the bottom body panel between the rear hips (2 indicators).

The indicators are angled to illuminate the ground in front, beneath, and to the sides of Spot. This is intended to maximize surface reflections and improve visibility from around corners, behind small obstacles, beneath grated surfaces such as stairs and walkways, and other directions where Spot might not be directly or easily visible.

The brightness and the intended maximum exposure to direct eyesight are in compliance with IEC 62471. To further reduce direct eye exposure, the indicators automatically power off when Spot tilts or rolls more than 90 degrees from the horizontal (neutral) position, and when it rolls onto its back.

Indicator location	**Max lux**[1]
Front panel	1030 lux
Front and rear hips	1010 lux

[1]Lux measured by light projected onto the floor when Spot is in the stand position.

2.8.2. Buzzer

The buzzer produces a tone of 5,333 Hz. The high pitch is intended to make the buzzer distinct from other common tones.

The buzzer volume is adjustable and can produce a maximum sound pressure level of 110 dBA at 1 meter in front of Spot.